Biodivers Conserv
DOI 10.1007/s10531-009-9597-3

ORIGINAL PAPER

Influence of harvest gaps and coarse woody material on click beetles (Coleoptera: Elateridae) in Maine's Acadian forest

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Received: 13 October 2007 / Accepted: 19 February 2009
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Abstract In order to maintain biodiversity in forests, it has been recommended that harvests be designed after patterns of natural disturbance. Using a long-term study that includes harvest treatments designed to emulate tree-fall gap disturbances in Maine's Acadian forest, we examined how the species richness, abundance, diversity, and assemblage similarity of click beetles inhabiting coarse woody material (CWM) were affected by gap harvesting and CWM characteristics (diameter, degree of decay, and type of wood). There were few differences in beetle assemblages between 0.07 and 0.12 ha harvest gap treatments. Four of the most common species had higher abundances under a closed forest canopy than within harvest gaps. Species richness and total abundance were higher in CWM that had larger diameters and were more decayed. Species assemblages also differed with the degree of wood decomposition. Diversity was higher in CWM from softwood trees than hardwood trees. Results from this study suggest that small (<0.2 ha) harvest gaps with living trees retained throughout the gap can maintain click beetle assemblages similar to that of an unharvested forest. Forest managers also need to address the temporal continuity of CWM, including different types of wood (hardwood and softwood), a range of decay conditions, and a range of diameter classes, especially larger diameters (>35 cm).

Keywords Click beetles · Conservation · Deadwood · Down woody debris · Elateridae · Forest management · Insect diversity · Logging · Ecologically based disturbance · Saproxylic insects

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Published online: 07 March 2009

Introduction

Maintaining biodiversity in forest ecosystems while meeting economic goals for wood production requires forest management that is based upon sound ecological principles. To achieve this goal, it has been recommended that forestry practices emulate patterns of natural disturbances (e.g., insect outbreaks, fire, and windthrow) using harvest design, variable retention of living trees, rotation schedule, and deadwood maintenance (Franklin and Forman 1987; Franklin 1989; Hansen et al. 1991; Hunter 1993; Seymour et al. 2002). Although there is a growing body of literature addressing these approaches (Haila et al. 1994; Bunnell 1995; Jonsell et al. 2005), there is still relatively little understanding of the influence of forest management on the majority of forest species (Komonen 2001; Grove 2002).

The Acadian forest is a transitional eco region between the boreal forest to the north and the eastern temperate forest to the south (Bailey 1995; Lorimer and White 2003). Large, stand-replacing disturbances were unusual in the pre-European settlement forest, with recurrence intervals of hundreds of years or more (Lorimer 1977; Seymour et al. 2002; Lorimer and White 2003; Seymour 2005). Disturbances occurred predominantly as small and frequent treefall gaps, resulting in a forest structure dominated by late-successional species in multi-aged stands. Therefore, a silvicultural system modeled after this natural disturbance regime would include gap creation and the retention of biological legacies, including deadwood and living trees that will not be harvested (Seymour et al. 2002; Franklin et al. 1997).

Deadwood, including snags and downed coarse woody material (CWM), supports a wide variety of animals, fungi, and plants and is vital for conserving biodiversity in forests (Speight 1989; Lindenmayer and Franklin 2002). Diameter of the CWM is important; usually larger diameters correspond to greater use or higher species richness and abundance (Gutzwiller and Anderson 1987; Jonsell et al. 1998; Kolström and Lumatjarvi 2000; Yee et al. 2001). Stage of decay also affects species assemblages as decomposing wood is occupied by a succession of species (Grove 2002). Tree species or wood type (hardwood or softwood) can affect saproxylic communities, but becomes less important with advancing stages of decay (Jonsell et al. 1998).

Harvesting can have substantial influence on a forest ecosystem, in some cases depleting the amount of deadwood over time through the removal of large living trees (potential deadwood), slash disposal, and site preparation (Fridman and Walheim 2000). Harvesting can also alter the distribution of size and decay class through the mechanical crushing of larger pieces in late-stage decay and through the addition of small diameter slash (Freedman et al. 1996; Fraver et al. 2002). In some intensively managed forests, many deadwood-dependent species, including fungi and invertebrates, are Red Listed or considered threatened (Lindenmayer et al. 1996; Ranius and Jansson 2000; Ehnström 2001; Tikkanen et al. 2007), and some species are now extinct (Kirby and Drake 1993).

Many species of click beetles (Coleoptera: Elateridae) live in forest soil and deadwood without causing serious damage to living trees, despite being generally regarded as pests in agricultural systems. In fact, elaterids are important in nutrient cycling (Wolters 1989), as predators of forest pests (Morris 1951; Yano et al. 1984), and as prey for birds and other forest biota (Barron and Walley 1983; Holmes and Robinson 1988; Heinrich and Bell 1995). The proportion of saproxylic species in North America is unknown, but 100 and 52% of *Ampedus* species in Central Europe and Great Britain are considered saproxylic, respectively (McLean and Speight 1993). Some saproxylic click beetle species are considered threatened or endangered (Anonymous 1999; Alexander 2003; Zach 2003), but the

deficiency of knowledge regarding most species (especially outside Europe) is an impediment to their conservation. Elaterids are important taxa to study because the family is species rich, numerically abundant, and the species have diverse food and habitat preferences.

We hypothesized that creating small canopy gaps modeled after natural disturbance patterns would retain adequate habitat structure for click beetles, thus maintaining elaterid diversity and abundance in managed forests. We tested whether the abundance and composition of click beetles living in CWM were affected by gap harvesting and CWM characteristics, including decay class, diameter, and wood type (softwood or hardwood).

Methods

Study site

This study took place in the Penobscot Experimental Forest in east-central Maine (44°50′N, 68°35′W). The dominant tree species include eastern hemlock [*Tsuga canadensis* (L.) Carr.], red (*Picea rubens* Sarg.) and white spruce [*P. glauca* (Moench) Voss], balsam fir [*Abies balsamea* (L.) Mill.], eastern white pine (*Pinus strobus* L.), northern white cedar (*Thuja occidentalis* L.), red maple (*Acer rubrum* L.), paper (*Betula papyrifera* Marsh.) and gray birch (*B. populifolia* Marsh.), and quaking (*Populus tremuloides* Michx.) and bigtooth aspen (*P. grandidentata* Michx.). The Penobscot Experimental Forest has a complicated history of insect outbreaks and repeated partial cuttings that resulted in multi-cohort stand structures of many species (R. Seymour, unpublished data). Soils range from glacial till ridges with well-drained or sandy loams to flat areas between ridges with poorly to very poorly drained loams and silt loams (Brissette 1996).

Sampling for this study used experimental plots established by the University of Maine's Acadian Forest Ecosystem Research Program, a long-term study comparing two silvicultural systems patterned after the natural disturbance regime of the Acadian forest (Saunders and Wagner 2005). The silvicultural systems include a 10 and 20% expanding-gap harvest system with permanent reserve trees. The 10% harvest treatment removes 10% of the canopy in harvest gaps on a 10-year cutting cycle with 10% of the overstory basal area permanently reserved. The 20% harvest treatment removes 20% of the canopy in harvest gaps on a 10-year cutting cycle with 10% of the overstory basal area permanently reserved. Target gap size was 0.1 ha for the 10% treatment and 0.2 ha for the 20% treatment. However, due to the need to include logging trails as part of the percentage canopy removal, the average gap size was slightly smaller than the initial target. Because the silvicultural prescription required working with the variation in stand conditions needing regeneration and understory release, the range of gap sizes also varied within each treatment area. Forty-five gaps were created in this study, with a mean gap size of 0.07 ha for the 10% treatment (range 0.01–0.13 ha) and 0.12 ha for the 20% treatment (range 0.03–0.22 ha).

Each treatment was replicated three times in 9.4–11.3 ha treatment plots using a randomized complete block design. The proportion of area in harvest gaps and access trails was 9.7% for the 10% treatment and 18.8% for the 20% treatment, thus closely matching the percentage canopy removal for each treatment. The first harvest for each block was applied in 1996, 1997, and 1998. At the time of this study, vegetation in the gaps had recovered for 3–6 years since the harvest.

Field invertebrate sampling

We compared click beetle assemblages between harvest treatments and among three characteristics of CWM: log diameter, wood type, and decay class. To test the influence of the harvest treatments on click beetles in CWM, we systematically searched all six harvest treatment plots for downed logs that met specific characteristics. These characteristics were stratified and included harvest treatment (10 or 20%), canopy condition (located in a harvest gap or under the adjacent closed canopy at least four meters from the nearest canopy opening), wood type (softwood or hardwood), two decay classes after Fraver et al. (2002), and a range of log diameters $\geq$14 cm.

We found that a four meter minimum distance from the nearest canopy opening was sufficient to constitute a closed canopy condition over downed logs. The use of reserve trees in the harvest gap treatments minimized the abrupt edge typically created in many harvest gap experiments. This assumption was later corroborated by our detecting statistical differences in the abundance of some species of click beetles in logs between closed canopy and harvest gap conditions.

Only logs in decay classes 2 and 4 (after Fraver et al. 2002) were selected for our study. Logs in Decay Class 2 had wood that was sound to somewhat rotten; bark may or may not have been attached; branch stubs were firmly attached but only larger stubs were present; and had retained a rounded shape and lay on duff. Logs in Decay Class 4 had wood that was mostly rotten, "fluffy" when dry and "doughy" when wet; branch stubs were rotted; bark was detached or absent (except *Betula*); and log was oval in cross section and usually substantially buried in duff.

Thus, 96 logs (2 harvest treatments $\times$ 2 canopy conditions $\times$ 2 CWM wood types $\times$ 2 decay classes $\times$ 2 samples per treatment plot $\times$ 3 blocks) were sampled. Due to logistical difficulties, data from one log were dropped, leaving 95 logs for analysis.

To sample beetles inhabiting each piece of CWM, we constructed emergence traps of dark mesh tents that enclosed a 1.0 m length of each log (Thomas 2007). For Decay Class 2 logs, the trap mesh was wrapped completely under the log. For Decay Class 4 logs, the mesh was tucked between the leaf litter and the log, without reaching beneath the highly decomposed material.

Beetles emerging from each log inside the tent were collected in one of two clear collecting bottles (upper or lower) containing propylene glycol that were positioned approximately 1.0 m above the log at the apex of each tent. We fitted the inverted upper collecting bottle with a funnel to catch stronger flying insects. The lower upright collecting bottle was positioned directly beneath the upper bottle to capture weaker flying insects that could not maneuver into the funnel of the upper bottle, including most species of click beetles.

Biweekly insect samples were collected from the traps from mid-June through mid-September of 2002. All insect samples were stored in 70% ETOH or pinned. Adult click beetle specimens were identified to species by S. L. Thomas and S. LaPlante and incorporated into the University of Maine Insect Collection.

Analytical approach

We used several approaches to examine assemblages and habitat characteristics. The principal response variables were species richness (number of species), total abundance (number of individuals), species abundance (number of individuals of the 15 most common species), diversity, and assemblage similarity. For log diameter, the diameters of the

smaller and larger ends (one meter apart) of each piece of CWM were averaged and used for all analyses.

We examined the relationships between the individual beetle species abundances with various habitat characteristics using general linear models (PROC GENMOD, SAS Institute Inc. 2000). A negative binomial distribution was used because the data were over-dispersed. Models were developed based on the main effects and all pairwise interactions, selecting an optimal model with the second order Akaike Information Criterion (Akaike 1974). Due to insufficient sample size for many insect species, only the 15 most common species of click beetles and false click beetles were included. We developed general linear models using harvest treatment, canopy condition, diameter, decay class, wood type, species, and their pairwise interactions as effects. In the general linear models, CWM diameter was used as a continuous variable.

To determine whether the size of a harvest gap around a sample log influenced the abundance of the emerging beetles, Thomas (2007) also developed a reduced model based on logs only in harvest gaps and using area of the gap as a main effect. Harvest treatment and gap area may exhibit colinearity, so these reduced models were generated excluding one or other of these two variables. Harvest treatment and gap area were not significant ($P > 0.05$) for any reduced model (Thomas 2007).

Because the general linear models were limited to the most abundant species, we used several other analyses (*t*-tests of richness and abundance, Jaccard similarity measures, rarefaction, and indicator species analysis) to examine the relationships between whole beetle assemblages and the different habitat characteristics. In order to compare species assemblages between classes of each habitat characteristic, we pooled log samples within each class and across all other habitat characteristics. For example, we compared all hardwood CWM with all softwood CWM, pooling data from all other log factors (canopy condition, diameter class, decay class), replicates (3), and treatments (10% removal, 20% removal). Because the treatment plots had few large (>35 cm) diameter CWM (Fraver et al. 2002) and because other studies have found these larger diameter CWM to be important (see Grove 2002), we separated diameter into the following three classes: 14.0–24.5 cm (54 traps), 24.6–34.5 cm (35 traps), and >34.6 cm (6 traps), and used these size classes for analysis of click beetle similarity measures, species richness, total abundance, diversity, and indicator species analysis.

We used *t*-tests to compare differences between the pooled groups of logs for beetle richness and total abundance. Differences were considered significant at $P = 0.05$. Cluster analysis of Jaccard percent similarity measures using internet-based software (Brzustowski 2002) was used to compare similarity of species assemblages between pooled groups of logs for the different habitat characteristics.

The classification scheme resulted in unequal, but biologically meaningful, sample size. Therefore, rarefaction, which includes both richness and abundance, was used to estimate the true species diversity. We used PAST (Hammer et al. 2001) to calculate the expected number of species derived from random subsamples of the total abundance, comparing species diversity between subsamples of similar sizes (Sanders 1968; Hurlbert 1971). Diversity between two pooled groups of logs was considered significantly different when respective 95% confidence intervals did not overlap.

We assessed the affinity of each species for different habitat characteristics with indicator species analysis (Dufrêne and Legendre 1997) using PC-ORD (McCune and Mefford 1999). Because this analysis combines faithfulness (present in all samples of a group of logs) and exclusiveness (never present in other groups) of a species to a particular habitat characteristic, we used it to determine the importance of different habitat characteristics to

individual species. Indicator values range from 0 (no indication to any group of logs) to 100 (perfect indication to a group of logs). A Monte Carlo *P*-value for each indicator value was determined as the proportion of 1,000 randomized trials with an indicator value equal to or exceeding the observed indicator value.

Results

Taxa collected

We collected 801 adults of 56 species, including two species (*Isorhipis obliqua* Say and *Fornax canadensis* Brown) of Eucnemidae, the false click beetles. Individuals from the *Dalopius* genus were treated as one species. The list of species and the number of specimens of each from each log are given in Thomas (2007). *Ctenicera triundulata* Randall, *Ampedus mixtus* Herbst, and *Ampedus semicinctus* (Randall) were the most common species, representing 19, 15, and 5% of the total abundance, respectively. Fourteen species were singletons, and 22 species were represented by only two to nine individuals.

Harvest treatment

Species richness, total abundance, and diversity of beetles did not differ between 10 and 20% harvest treatments (Table 1). However, assemblage similarity, a measure of the overlap of species between habitats, was only 60%, indicating a difference in species composition between the two harvest treatments. *Agriotes stabilis* LeC. was a significant indicator of the 20% harvest treatment (Table 2). The mean abundances of the 15 most common species did not differ significantly between 10 and 20% harvest treatments ($P = 0.405$, Table 3).

Canopy condition

Species richness, total abundance, and diversity were not significantly different between closed canopy and harvest gap conditions (Table 1). *Ctenicera hieroglyphica* Say and *Limonius aeger* LeC. were significant indicators of closed canopy conditions (Table 2). Assemblage similarity was only 58%. Among the 15 most common species, the abundance of individual species differed between the two canopy conditions ($P = 0.001$). No species were more likely to be found in harvest gaps, but four species (*C. hieroglyphica*, *Ctenicera propola* LeC., *Dalopius* spp. Brown, and *L. aeger*) were more abundant under a closed canopy than in harvest gaps (Table 3; Fig. 1a).

Coarse woody material characteristics

Wood type

Species richness and total abundance were not significantly different between hardwood and softwood CWM; however, species diversity was higher in softwood CWM (Table 1). No species were indicators of softwood or hardwood logs (Table 2), but assemblage similarity was 64%, suggesting a difference in species composition. The mean abundance of the 15 most common species was not significantly different between the softwood and hardwood CWM ($P = 0.177$, Table 3).

Table 1 Summary of richness (number of species per log), total abundance (number of individuals per log), and rarefaction-estimated species diversity of click beetles

Source	Number of logs	Species richness[a]	Total abundance[b]	Diversity[d]
Harvest treatment				
10% removal	47	4.7 (2.7)	7.9 (7.0)	36.9 ± 1.8 (355)
20% removal	48	4.8 (3.2)	9.0 (8.4)	35.0 ± 0.2 (355)
Type				
Hardwood	47	4.9 (3.5)	9.3 (9.2)	45.4 ± 2.3 (380)
Softwood	48	4.6 (204)	7.6 (5.9)	53.8 ± 0.8 (380)
Canopy condition				
Gap	47	4.5 (2.4)	8.0 (5.9)	55.9 ± 0.7 (420)
Closed canopy	48	5.0 (3.4)	8.9 (9.2)	55.1 ± 1.7 (420)
Deacay glass				
Decay glass 2	48	3.9 (2.7)	6.6 (8.2)	55.9 ± 0.6 (360)[e]
Decay glass 4	47	5.6 (2.9)[c]	10.3 (6.7)[c]	43.5 ± 3.5 (360)
Diameter class				
Small vs. Medium				
14–24.75 cm	54	4.2 (7.3)	6.7 (47.9)	39.0 ± 2.9 (270)[e]
25–34.75	35	4.7 (2.6)	8.5 (6.5)	31.0 ± 0.2 (270)
Medium vs. Large				
25–34.75 cm	35	4.7 (2.6)	8.5 (2.6)	26.3 ± 3.2 (110)[e]
35 cm+	6	10.0 (4.3)[c]	24.0 (15.0)[c]	17.0±0.4 (110)

[a] Mean number of species per log (standard deviation)

[b] Mean number of individuals per log (standard deviation)

[c] Richness or abundance is significantly higher in this class as determined by *t*-test ($P = 0.05$)

[d] Rarefaction-estimated number of species ± confidence limits (number of individuals in the final random subsample of the total abundance)

[e] Diversity is significantly higher in this class (95% confidence limits)

Decay class

Species richness and total abundance were 46 and 56% higher in more decayed CWM (Decay class 4), respectively, but species diversity was higher in less decayed (Decay class 2) material (Table 1). *C. triundulata* was a significant indicator of Decay class 4 (Table 2), and the species composition was quite different (56% similarity). The abundance of individual species varied between the two decay classes ($P < 0.001$). Three species (*A. semicinctus*, *A. scapullaris*, and *M. castanipes*) were more abundant in less decayed CWM, and six species (*A. brightwelli*, *Athous orvus* Becker, *C. hieroglyphica*, *C. triundulata*, *Dalopius* spp., and *L. confusus*) were more likely to be found in more decayed CWM. Abundance of six species did not differ between decay classes (Table 3; Fig. 1b).

Diameter

Click beetle species richness was 137 and 115% higher in the large diameter logs than in small and medium diameter logs, respectively (Table 1). Total abundance was 260 and 183% higher in the large diameter class than in the small and medium diameter classes,

Table 2 Indicator species analysis of click beetle species

Source	Species	Total number collected	Indicator value	*P*
Harvest treatment				
10% removal	No species			
20% removal	*Agriotes stabilis*	35	34.4	0.002
Type				
Hardwood	No species			
Softwood	No species			
Canopy condition				
Gap	No species			
Closed canopy	*Ctenicera hieroglyphica*	25	27.1	0.001
	Limonius aeger	24	27.3	0.001
Deacay glass				
Decay glass 2	No species			
Decay glass 4	*Ctenicera triundulata*	151	70.7	0.001
Diameter class				
14–24.75 cm	No species			
25–34.75 cm	No species			
35 cm+	*Agriotes collaris*	8	44.4	0.002
	Ampedus mixtus	121	35.7	0.097
	Athous brightwelli	14	27.8	0.020
	Athous rufifrons	35	64.0	0.001
	Ctenicera hieroglyphica	25	53.0	0.003
	Ctenicera triundulata	151	49.0	0.021
	Dalopius species	33	37.6	0.009
	Limonius aeger	24	39.4	0.013
	Limonius confusus	36	55.5	0.002

Only significant species ($P = 0.1$) with an indicator value of at least 25 are shown

Table 3 General linear model results of click beetle species abundance

Source	*df*	Chi-square	Pr>Chisq
Harvest treatment	1	0.69	0.405
Site within harvest treatment	4	24.72	<0.001
Beetle species	14	204.06	<0.001
Type (Softwood vs. Hardwood)	1	1.83	0.177
Diameter	1	29.8	<0.001
Canopy condition	1	11.06	0.001
Decay class	1	11.29	0.001
Beetle species * Canopy condition	14	48.19	<0.001
Beetle species * Decay class	14	77.29	<0.001

respectively. Conversely, species diversity decreased with increasing CWM diameter (Table 1). Nine species (*Agriotes collaris* LeC., *A. mixtus*, *A. brightwelli*, *Athous rufifrons* Randall, *C. hieroglyphica*, *C. triundulata*, *Dalopius* spp., *L. aeger*, and *L. confusus*) were indicators of the largest diameter class, while no species were indicators of the two smaller

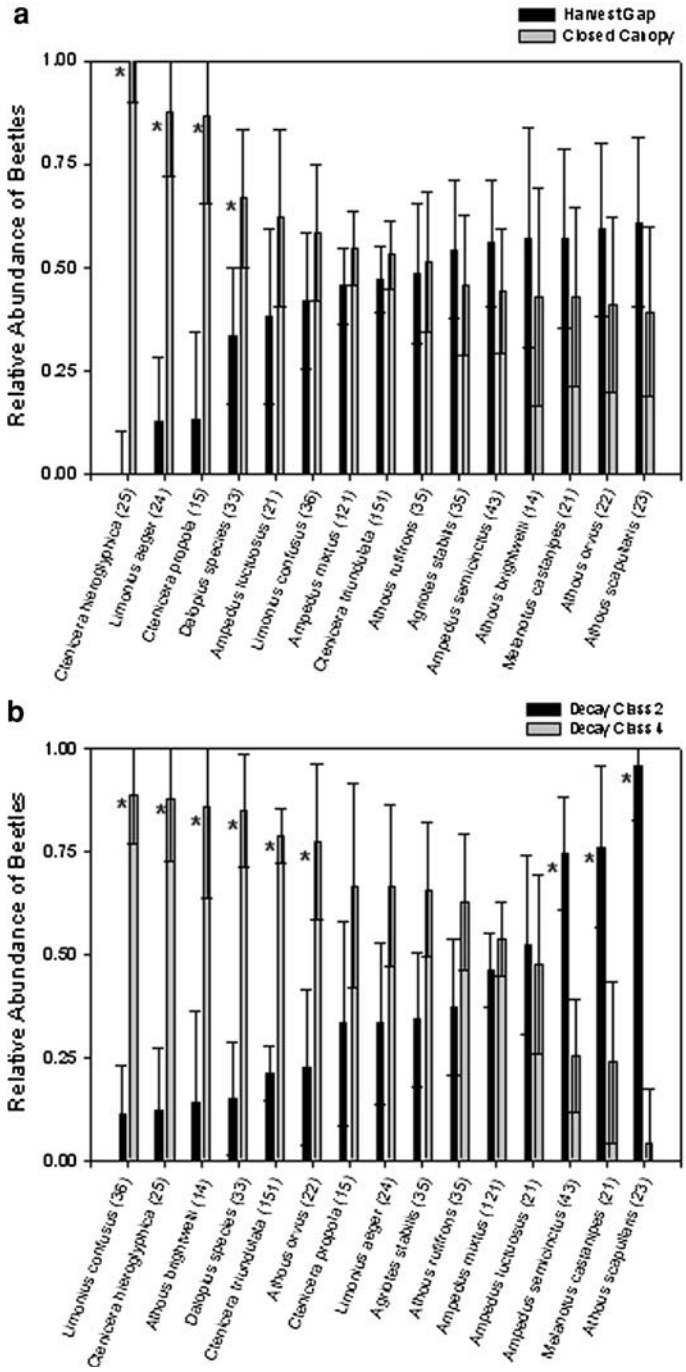

Fig. 1 Relative abundance of beetle species for **a** harvest gap and closed canopy and **b** decay class 2 and decay class 4. Individual species abundances in parentheses

classes (Table 2). The two smaller classes were 67% similar in species assemblage, but differences between the larger diameter class and the middle and smallest classes were greater at 47 and 46% assemblage similarity, respectively. The mean abundance of the 15 most common species increased with diameter of CWM ($P < 0.001$, Table 3).

Discussion

Harvest gaps

Gap size and harvest treatment

We found that size of the harvest gap did not influence click beetle abundance in CWM, and there were few differences between the 10 and 20% harvest treatments. Species richness, diversity, total abundance, and individual abundances of the 15 most common species did not differ significantly, and only *A. stabilis* was an indicator of the 20% harvest treatment. However, there were differences in species composition between the harvest treatments (40% dissimilarity), mostly because of species with fewer than six specimens. We do not know whether these uncommon species were ubiquitous or true representatives of their habitats. There are several possible explanations for the general lack of response. First, the range of harvest gap sizes examined was limited (0.01–0.22 ha). Second, the maximum area of the harvest gaps was small, so that even the largest gaps were still influenced by the surrounding stand. However, an experiment in a bottomland hardwood forest in South Carolina that created larger and a wider range of harvest gap sizes (0.13, 0.26, and 0.50 ha) revealed no difference in the abundance of insect herbivores (Ulyshen et al. 2005) or carabid predators (Ulyshen et al. 2006). Third, the living permanent reserve trees that were distributed throughout the harvest gaps (6–21% of starting stand basal area) in this study may have moderated the influence of the canopy openings.

Canopy condition

Many saproxylic species can live in sun-exposed deadwood (Jonsell et al. 2004; Lindhe and Lindelöw 2004); some species specialize in these environments (Kaila et al. 1997). However, other species require or are more abundant in shaded deadwood (Hilszczanski et al. 2005), especially those species living in wood in the later stages of decay (Jonsell et al. 1998).

The harvest gaps influenced click beetles species assemblages. The species composition differed, with 42% of the species found exclusively in either harvest gaps or under the closed canopy, a lower similarity compared to the harvest treatments. Four of the 15 most common species were found in greater abundances in CWM under a closed canopy, and two of these species (*C. hieroglyphica* and *L. aeger*) were strongly associated with the closed canopy. None of the most common species had higher abundances in harvest gaps, and no species were indicators of harvest gaps. Our results, however, were not as dramatic as differences found in studies examining saproxylic arthropods on exposed deadwood in clearcuts (e.g., Kaila et al. 1997). This contrast was not surprising, as the relatively small harvest gaps for both treatments (0.07 and 0.12 ha) had substantial numbers of living trees retained throughout the gaps that reduced sun exposure of the CWM sampled in this study.

Coarse woody material characteristics

Speight (1989) defined saproxylic insects as those that are "dependent, during some part of their life cycle, upon the dead or dying wood of moribund or dead trees (standing or fallen), or upon wood-inhabiting fungi, or upon the presence of other saproxylics." Deadwood-dependent beetles outnumber all terrestrial vertebrates by 2-to-1 (Parker 1982; Hunter 1999), and saproxylics comprise a substantial percentage of forest insects (Martikainen et al. 2000; Siitonen 2001). Saproxylic insects respond to a range of deadwood characteristics, including those assessed in this study.

Wood type

Recently dead wood is often colonized by insect species with narrow host specificity, but as decomposition advances, there is less host specificity to individual tree species, although there usually remains differentiation in saproxylic species assemblages between hardwood and softwood CWM (Jonsell et al. 1998; Kolström and Lumatjarvi 2000; and Wikars 2002). In this study, however, we found no differences in species richness or total abundance between hardwood and softwood CWM. Nevertheless, softwood CWM had higher diversity than hardwood CWM, and 36% of the species were found exclusively in either softwood or hardwood CWM. These results may be due to our use of CWM in later stages of decay, where host specificity is less likely, and to the possibility that other CWM characteristics were more important in determining species assemblage than wood type.

Decay class

A succession of insect species colonizes wood as it decays. Fresh CWM (e.g., Decay Class 1) is inhabited by a distinct fauna that often require fresh phloem or sapwood (Hammond et al. 2004). As wood decomposes (Decay Classes 2 and 4), assisted by a series of fungal species, the species richness of fungivores, scavengers, and predators increases in response to increasing microhabitat diversity (Siitonen 2001). Our study recognized this succession by deliberately avoiding sampling the earliest stages of decay, as Elateridae are not common in fresh deadwood. We nevertheless found differences in click beetle colonization of CWM between middle and later stages of decay (Decay Class 2 and 4).

In this study, species richness and total abundance were higher in more decomposed CWM (Decay Class 4). One species (*C. triundulata*) was strongly associated with Decay Class 4. Three of the 15 most common species were more abundant in Decay Class 2, and six of the most common species were more abundant in Decay Class 4. Assemblage similarity was lower than between wood types (57%), suggesting a difference in species composition between decay classes. Although few studies have examined non-pest species, it is assumed that most click beetle species occupying later decay class CWM are scavengers and predators (Gur'yeva 1969), following the basic trend found by Siitonen (2001). In European forests, where arthropods have been studied more thoroughly, the preponderance of rare and endangered invertebrates is found in the later stages of decomposing wood (Ehnström 2001). The lack of available CWM and the reduced duration of later stages of decomposition due to CWM crushing during harvest (Freedman et al. 1996), suggests that available habitats for North American forest invertebrates requiring downed wood in later stages of decomposition could be reduced.

Diameter

The size of the deadwood is important to most saproxylic species. Some species are able to use a wide range of CWM sizes, but many species seem to rely on large-sized logs. As a result, there tended to be different species assemblages in logs of increasing sizes. A positive relationship between CWM diameter and species richness and abundance has been found in other studies (Jonsell et al. 1998; Grove 2002; Hammond et al. 2004; but see Araya 1993). In this study, although click beetle diversity decreased with increasing CWM diameter, all other measures increased with increasing log diameter. Abundance of the 15 most common species increased with log diameter. Overall species richness and total abundance of click beetles were higher in the largest diameter class. Assemblage similarity was low, with 53–54% of the species found exclusively in either the largest diameter class or the two smaller classes; and was the lowest similarity found among habitats in this study. Further, no species were strongly associated with smaller diameter logs, but nine species, including *A. mixtus*, were indicators of larger diameter logs (>35 cm). Hammond et al. (2004) also listed *A. mixtus* as an indicator of larger diameter (>41 cm) logs. These results suggest that these species prefer or may be dependent on larger diameter logs. In Europe, a number of click beetle species that are listed as endangered or vulnerable live in large diameter CWM, snags, or hollow trees (Anonymous 1999; Ranius and Jansson 2000, 2002; Zach 2003). However, not all species seem to respond to diameter. For example, *Melanotus castinipes* abundance was not correlated with diameter, as was also found by Ranius and Jansson (2000).

Other factors being equal, larger diameter trees and CWM generally decay more slowly, providing a more stable microclimate that benefits particular invertebrate species (Grove 2002). For example, the endangered click beetle *Limoniscus violaceus* Müller appears to require decaying wood with a certain moisture content to survive (Zach 2003). Larger diameter CWM can also maintain a greater diversity of fungi, including those species that are specific to larger wood pieces (Grove 2002). Click beetles are commonly assumed to be generalists, presumably increasing in numbers with increasing prey abundance, but there are several species with known specific prey or habitat requirements. For example, *Elater ferrugineus* is habitually found where its prey, *Dorcus parallelipedus* (Coleoptera: Lucanidae), is active, although it is also known to prey on the threatened scarab beetle *Osmoderma eremita* (Svensson et al. 2004). Furthermore, several endangered species in Europe are known to have habitat requirements specific to wood decomposed by certain fungus species (Anonymous 1999). Although we did not assess the fungi, it is likely that some beetle species we collected also respond to specific rot types or prey species that were dependent on particular fungal species or CWM diameters.

Conclusion

Creating harvest gaps that were relatively small (<0.2 ha) and included a substantial number of living trees and CWM permitted click beetle assemblages to vary little between harvest gaps and closed canopy; although several individual species were associated with the closed canopy. A number of studies indicate that Elateridae, as do many other saproxylic fauna, use CWM with a wide variety of characteristics. Because there are such wide-ranging preferences within Elateridae, each of the CWM characteristics studied here appeared to be important for maintaining biodiversity. Thus, forest managers should address the temporal continuity of deadwood (decomposition states), different wood types

(hardwood and softwood), and a range of diameter classes (especially diameters >35 cm) in the Acadian forest.

Acknowledgments We gratefully acknowledge the staff at the Maine Forest Service for access to their insect collection and the staff at the Penobscot Experimental Station. We express our sincere gratitude to Edward C. Becker and Serge Laplante for correcting our identifications of the click beetle reference collections. We also thank Steve Woods. This research was made possible by funding from the USDA National Research Initiative Competitive Grants Program (Award No. 00-35101-9239) and Maine Agricultural and Forestry Experiment Station (MAFES Publication 3011).

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